

Quantum Computing in the T0 Framework: Theoretical
Foundations and Experimental Predictions Proof of ϕ -QFT
Equivalence with Bell-Corrected Entanglement

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Abstract

We examine a formulation of the quantum Fourier transform within the T0 time-mass duality theory. We examine whether the ϕ -hierarchical quantum Fourier transform (ϕ -QFT) fulfils the same function as the standard QFT for period finding in Shor's algorithm; the proof does not hold in its present form, and an advantage over the standard QFT is not demonstrated. Three mechanisms are examined: (1) energy field superposition as a deterministic alternative to probabilistic collapse, (2) correlation fields as a description of the Bell correlations, where the fractal damping establishes no local bound and the Bell violation remains, and (3) fractal damping that suppresses decoherence. The theory makes precise theoretical predictions; the central one is a CHSH deviation of order $\xi \sim 10^{-5}$ from the Tsirelson bound. Hardware measurements (May 2026, IBM Heron r2) confirm clear Bell violation ($S = 2.74$, 96.9% of the Tsirelson bound) and the runnability of T0 circuits; the ξ -effect itself lies far below current NISQ noise and is not resolvable with present hardware. We provide a complete Python implementation demonstrating 100 percent success rate on benchmark factorizations up to $N=143$. The finding is therefore twofold: the Bell violation is confirmed and the circuits run; the characteristic ξ prediction is not testable with current hardware.

.1 Introduction

Motivation and Context

The standard quantum computing paradigm faces fundamental conceptual challenges: the measurement problem, apparent nonlocality in entanglement, and the lack of a deterministic underlying framework. The T0 time-mass duality theory [1], based on the fundamental relation $T(x,t) \cdot E(x,t) = 1$ and the universal parameter $\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4}$, offers an alternative perspective that addresses these problems while remaining compatible with experimental quantum mechanics.

Main Contributions

This work establishes:

1. **Theoretical equivalence:** Proof attempt that ϕ -hierarchical QFT reproduces the period finding of standard QFT (Theorem .3.4); in its present form the proof does not hold
2. **Bell corrections:** Mathematical framework for Bell test modifications with predicted deviations in multi-qubit systems that are not resolvable with present NISQ hardware (Section .4)
3. **Stability improvement:** Demonstration that ξ -damping provides natural decoherence suppression (Corollary .3.5)
4. **Experimental protocols:** Detailed predictions for 73-qubit Bell tests, testable only with error-corrected hardware; the ansatz of a spatial correlation delay for satellite experiments is not a prediction of FFGFT (Section .4)
5. **Implementation:** Complete algorithmic implementation with verified performance (Section .8)

Structure

Section .2 summarizes T0 foundations. Section .3 presents the central theoretical results. Section .4 develops Bell test modifications. Section .5 applies the framework to Shor's algorithm. Section .4 details experimental predictions. Section .8 describes the Python implementation.

.2 T0 Framework Foundations

Core Principles

Definition .2.1 (T0 Time-Mass Duality). The fundamental relation of T0 theory is:

$$T(x,t)(x,t) \cdot E(x,t)(x,t) = 1 \quad (1)$$

where $T(x,t)$ is the dynamic time field and $E(x,t)$ is the energy density field.

Definition .2.2 (Universal Parameters). The T0 framework is characterized by:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \quad (\text{coupling strength}) \quad (2)$$

$$\phi_{\text{par}} = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad (\text{golden ratio}) \quad (3)$$

$$\Delta f = 3 - \xi \approx 2.9999 \quad (\text{fractal dimension}) \quad (4)$$

Energy Field Qubits

Unlike standard qubits, which are represented as complex vectors $\alpha|0\rangle + \beta|1\rangle$ in Hilbert space, T0 qubits are described by energy field configurations in cylindrical coordinates.

Definition .2.3 (T0 Qubit). A T0 qubit is characterized by the triple (z, r, θ) where:

- $z \in [-1, 1]$: projection onto the computational basis axis ($z = 1 \Leftrightarrow |0\rangle$)
- $r \in [0, 1]$: superposition amplitude (radial distance from the z-axis)
- $\theta \in [0, 2\pi)$: phase (azimuthal angle)

with normalization condition $z^2 + r^2 = 1$.

Remark .2.4. The crucial conceptual shift: r^2 is *not* a probability, but represents *energy density* of the superposition state. This enables deterministic evolution while maintaining quantum mechanical interference.

Geometric Foundation: Torus Structure and Numerical Accuracy

Although T0 qubits are represented in cylindrical coordinates (z, r, θ) for computational efficiency, the underlying physical structure lies in a **toroidal energy vortex** with fractal dimension $\Delta f = 3 - \xi$.

The cylindrical representation is a **local approximation** valid when the toroidal major radius $R \gg r$ (tube radius). For $R \rightarrow \infty$, the torus locally approaches a cylinder:

$$\text{Torus}(R \rightarrow \infty) \xrightarrow{\text{local}} \text{Cylinder}(z, r, \theta)$$

For quantum systems at the proton scale, the aspect ratio is enormous:

$$\frac{R}{r} \sim 2.5 \times 10^{18} \quad (\text{proton scale})$$

This extreme ratio makes the cylindrical approximation **exact in the limit** while maintaining optimal computational efficiency.

Accuracy Analysis:

Comprehensive numerical simulations comparing cylindrical, toroidal, and hybrid approaches show excellent agreement for large aspect ratios:

Table 1: CHSH parameter comparison: 73-qubit system

Method	CHSH value	Δ vs. IBM	Relative error (%)
Standard QM	2.828427	8.84×10^{-2}	3.23
IBM measured (28 May 2026)	2.74	—	—
T0 Cylindrical	2.827888	8.79×10^{-2}	3.21
T0 Toroidal	2.827943	8.79×10^{-2}	3.21
T0 Hybrid	2.828027	8.80×10^{-2}	3.21

All methods lie equally far ($\approx 3.2\%$) above the measured value $S = 2.74$ (Section .7); the measurement does not distinguish between them.

Key Findings:

- **Cylindrical optimality:** For $R/r > 10^{12}$, cylindrical calculations provide optimal accuracy with $O(n^2)$ computational complexity
- **Convergence:** All physically consistent methods converge to within 0.02 percent for proton-scale ratios

- **Computational efficiency:** The cylindrical representation enables a speedup by a factor n ($O(n^2)$ vs. $O(n^3)$) for multi-qubit systems; this is a polynomial, not an exponential gain

Physical Implementation:

The toroidal geometry is implemented through physically consistent corrections that respect fundamental limits:

1. **Non-singular curvature:** Exponential correction factor

$$\alpha = \exp\left(-\frac{\xi}{\sqrt{R/r}}\right) \approx 1 \quad \text{for } R/r > 10^{12}$$

2. **Energy conservation:** Normalization factor bounded to $[0.999, 1.001]$, ensuring physical consistency

3. **Fractal dimension:** All corrections respect the $\Delta f = 3 - \xi$ boundary condition

Physical Implications:

The cylindrical approximation successfully captures all essential T0 features:

1. **Bell damping preservation:** The fractal damping factor $\exp(-\xi \ln(n)/\Delta f)$ arises from torus geometry and is exactly preserved in cylindrical coordinates
2. **Charge quantization:** The quantization of electric flux through the torus hole reduces for $R/r \rightarrow \infty$ to phase quantization $\theta_k = 2\pi k / \phi_{\text{par}}^m$ in cylindrical coordinates
3. **Spin representation:** Winding numbers (n_ϕ, n_θ) on the torus are bijectively mapped to spin states $|\uparrow\rangle, |\downarrow\rangle$
4. **Computational efficiency:** $O(n^2)$ quantum gate operations vs. $O(n^3)$ for complete toroidal calculations

Optimal Method Selection by Aspect Ratio:

Table 2: Recommended approach by system scale

Aspect ratio	System type	Optimal method	Accuracy gain
$R/r < 10^6$	Macroscopic rings	Toroidal	Up to 85 percent
$10^6 \leq R/r \leq 10^{12}$	Mesoscopic	Hybrid	~0.1 percent
$R/r > 10^{12}$	Atomic/proton	Cylindrical	—

Transition to Quantum Computing:

For practical implementation of quantum algorithms at atomic scale ($R/r > 10^{12}$), we use the cylindrical representation with torus-derived parameters:

$$\text{Bell damping: } \mathcal{D}(n) = \exp\left(-\frac{\xi \ln(n)}{\Delta f}\right) \quad (5)$$

$$\text{Phase quantization: } \theta_k = \frac{2\pi k}{\phi_{\text{par}}^m}, \quad k, m \in \mathbb{Z} \quad (6)$$

$$\text{Energy normalization: } z^2 + r^2 = 1 \quad (7)$$

$$\text{Torus parameter: } \alpha = \exp\left(-\frac{\xi}{\sqrt{R/r}}\right) \approx 1 \quad (8)$$

This approach preserves the **conceptual foundation** of toroidal FFGFT geometry while providing **practical efficiency** for scalable quantum computations.

Remark .2.5 (Geometric Hierarchy). The complete geometric description follows a three-level hierarchy:

1. **Fundamental:** Toroidal energy vortex with fractal dimension $\Delta f = 3 - \xi$
2. **Effective:** Cylindrical T0 qubits with Bell damping and torus parameters
3. **Computational level:** Quantum gates and algorithms (Shor, Grover, etc.)

The cylindrical representation provides the optimal bridge between levels 1 and 3, preserving all essential physical properties while enabling efficient computation.

When does torus geometry matter?

Hypothesis: Toroidal corrections become significant only for $R/r < 10^6$.

Test systems:

- **Superconducting ring qubits:** $R \sim 10 \mu\text{m}$, $r \sim 1 \mu\text{m} \Rightarrow R/r \sim 10$
 - Predicted improvement: ~85 percent accuracy gain with toroidal calculations
 - Testable with current SQUID technology
- **Graphene torus structures:** $R \sim 1 \text{ nm}$, $r \sim 0.1 \text{ nm} \Rightarrow R/r \sim 10$
 - Predicted improvement: ~80 percent accuracy gain
 - Fabrication through carbon nanotube manipulation
- **Molecular ring qubits:** Cyclodextrin or similar $\Rightarrow R/r \sim 5\text{--}10$
 - Maximum toroidal effects expected
 - Potential for room-temperature quantum computing

Prediction: For $R/r > 10^{12}$ (all atomic systems), cylindrical and toroidal calculations agree to within <0.02 percent, confirming the validity of the cylindrical approximation for quantum computing.

Numerical Implementation:

The complete source code for the toroidal vs. cylindrical analysis, including the formulations that avoid numerical instabilities, is available at:

<https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/python/>

All calculations respect physical bounds:

- Bell correlations: $E(a, b) \in [-1, 1]$
- CHSH parameter: $S \in [0, 2\sqrt{2}]$
- Torus corrections: $\alpha \in [0.999, 1.001]$ for $R/r > 10^{12}$

Summary:

For quantum computing applications where $R/r > 10^{12}$ (all practical scenarios), the cylindrical representation is:

- **Physically exact:** Equivalent to torus geometry in the appropriate limit
- **Computationally optimal:** $O(n^2)$ vs. $O(n^3)$ operations
- **Numerically stable:** No singularities or convergence problems
- **Hardware-tested:** clear Bell violation measured ($S = 2.74$, 96.9% of Tsirelson); the ξ -level deviation is below current NISQ noise

$\Rightarrow$ Recommended implementation for all T0 quantum computing applications at atomic scale.

For future experiments with macroscopic qubits ($R/r < 10^6$), complete toroidal calculations may offer significant accuracy improvements and should be considered.

Modified Quantum Gates

Proposition .2.6 (T0 Hadamard Gate). *The T0 Hadamard gate with Bell damping for an n -qubit system is:*

$$H_{T0}^{(n)} : (z, r, \theta) \mapsto \left(r \cdot e^{-\xi \ln(n)/\Delta f}, z \cdot e^{-\xi \ln(n)/\Delta f}, \theta + \frac{\pi}{2} \right) \quad (9)$$

Proof. The transformation $(z, r) \rightarrow (r, z)$ implements basis change. The exponential factor $\exp(-\xi \ln(n)/\Delta f)$ represents Bell damping, which stabilizes multi-qubit entanglement (see Section .4). $\square$

.3 Main Theoretical Results

ϕ -Hierarchical Quantum Fourier Transform

Definition .3.1 (ϕ -QFT). The ϕ -hierarchical QFT on n qubits applies phases $2\pi/\phi_{\text{par}}^k$ instead of $2\pi/2^k$:

$$\phi\text{-QFT} : |x\rangle \mapsto \frac{1}{\sqrt{Q_{\phi_{\text{par}}}}} \sum_{y=0}^{Q_{\phi_{\text{par}}}-1} e^{2\pi i xy / Q_{\phi_{\text{par}}}} |y\rangle \quad (10)$$

where $Q_{\phi_{\text{par}}} = \phi_{\text{par}}^n$ (compared to $Q = 2^n$ for standard QFT).

Period Finding Compatibility

Lemma .3.2 (ϕ -Coverage of Periods). *For every period $r \in [2, N]$ with $N < 2^{20}$, there exists $k \in \mathbb{Z}$ such that:*

$$|r - \phi_{\text{par}}^k \cdot c| < \epsilon \quad (11)$$

for a rational number c with small denominator and $\epsilon < 1/(2r^2)$.

Proof. Consider the sequence $\{\phi_{\text{par}}^k\}_{k=0}^{\infty}$. Since $\phi_{\text{par}} \approx 1.618$, we have:

$$\phi_{\text{par}}^k = \phi_{\text{par}}^{k-1} + \phi_{\text{par}}^{k-2} \quad (\text{Fibonacci recurrence}) \quad (12)$$

The ratios $\phi_{\text{par}}^{k+1}/\phi_{\text{par}}^k = \phi_{\text{par}}$ are irrationally distributed. By Weyl's equidistribution theorem, for each r in a finite range, the fractional parts $\{\phi_{\text{par}}^k \bmod r\}$ are uniformly distributed modulo r .

For $N < 2^{20}$, we need $k \leq \log_{\phi_{\text{par}}}(N) \approx 20/\log_2(\phi_{\text{par}}) \approx 28.8$, i.e. $k \leq 29$. In this range:

- $\phi_{\text{par}}^1 = 1.618 \approx 2$
- $\phi_{\text{par}}^2 = 2.618 \approx 3$
- $\phi_{\text{par}}^3 = 4.236 \approx 4$
- $\phi_{\text{par}}^4 = 6.854 \approx 7$

A k with $|\phi_{\text{par}}^k - r| < 0.5$ exists for $r \in [2, 100]$ only at $r = 2, 3, 4, 7, 11, 18, 29, 47, 76$ (the Lucas numbers), not for every r , e.g. for $r = 5$, $\min_k |\phi_{\text{par}}^k - 5| = 0.764$. Moreover, the continued fraction algorithm is stable only under perturbations smaller than $1/(2r^2)$, not under perturbations up to 0.5. Coverage of all periods is therefore not proven by this argument. For $r = 2, \dots, 100$ only 9 of 99 values satisfy $|\phi_{\text{par}}^k - r| < 0.5$, and the lemma requires $\epsilon < 1/(2r^2) \leq 0.125$. $\square$

Bell-Enhanced Peak Detection

Lemma .3.3 (Bell Damping Effect). *With Bell-corrected phases, the QFT output satisfies:*

$$|\psi_{T0}\rangle = \frac{1}{Q} \sum_{k,y} e^{2\pi i k r y / Q_{\phi_{\text{par}}}} \cdot e^{-\xi |k r y / Q_{\phi_{\text{par}}} - m|^2 / \Delta f} |y\rangle \quad (13)$$

where $m = \text{round}(k r y / Q_{\phi_{\text{par}}})$.

Proof. The Bell correction factor (derived in Section .4) is:

$$\mathcal{D}_{\text{Bell}}(\theta) = \exp\left(-\xi \frac{\theta^2}{\pi^2 \Delta f}\right) \quad (14)$$

For phase differences $\Delta\phi = 2\pi k r y / Q_{\phi_{\text{par}}}$, the nearest integer is m . Damping suppresses contributions where $\Delta\phi$ deviates significantly from an integer multiple of 2π , i.e., off-peak components.

This enhances the correct peak at $y \approx Q_{\phi_{\text{par}}} / r$ while suppressing noise peaks, thus acting effectively as a filter. $\square$

Main Theorem

Theorem .3.4 (ϕ -QFT Equivalence for Period Finding). *For Shor's algorithm factoring $N < 2^{20}$ with error probability $\delta < 10^{-6}$:*

$$P_{\text{success}}(\text{Standard-QFT}) \leq P_{\text{success}}(\phi\text{-QFT}) \leq P_{\text{success}}(\text{Standard-QFT}) + \xi \quad (15)$$

Proof. We prove this in three steps:

Step 1: Period detection. By Lemma .3.2, for every period r dividing N :

$$\exists k : \left| \frac{Q_{\phi_{\text{par}}}}{r_{\phi_{\text{par}}}} - \frac{Q}{r} \right| < \frac{0.2Q}{r} \quad (16)$$

where $r_{\phi_{\text{par}}} = r \cdot \phi_{\text{par}}^k / 2^k$ for optimal k .

Step 2: Continued fraction stability. The continued fraction algorithm extracts r from the measured phase y/Q under the condition:

$$\left| \frac{y}{Q} - \frac{s}{r} \right| < \frac{1}{2r^2} \quad (17)$$

For $r < \sqrt{N}$ (which holds for useful periods), our perturbation from Step 1 does not satisfy this condition:

$$\frac{0.2Q}{r} = \frac{0.2 \cdot 2^n}{r} \gg 1 > \frac{1}{2r^2} \quad (18)$$

since $2^n \approx 2N \gg r$. Even in the relative form $0.2/r < 1/(2r^2)$ the inequality holds only for $r < 2.5$, i.e. only for $r = 2$. Continued fraction stability is therefore not established for general r , and the proof of the theorem does not hold at this point. Example $r = 3$: $0.067 > 0.056$.

Step 3: Bell enhancement. By Lemma .3.3, Bell damping increases the signal-to-noise ratio:

$$\text{SNR}_{\phi\text{-QFT}} = \text{SNR}_{\text{standard}} \cdot \left(1 + \frac{\xi \ln(r)}{\Delta f}\right) \quad (19)$$

For typical periods $r \in [2, 100]$:

$$\frac{\xi \ln(r)}{\Delta f} \approx \frac{1.333 \times 10^{-4} \times 4.6}{2.9999} \approx 2 \times 10^{-4} \quad (20)$$

This small improvement ensures:

$$P_{\text{success}}(\phi\text{-QFT}) \geq P_{\text{success}}(\text{Standard-QFT}) \quad (21)$$

The upper bound $P_{\text{success}}(\phi\text{-QFT}) \leq P_{\text{success}}(\text{Standard-QFT}) + \xi$ follows from the fact that $\phi\text{-QFT}$ cannot exceed perfect success and additional errors are bounded by ξ through perturbation analysis. $\square$

Corollary .3.5 (Decoherence Suppression). *Under phase noise $\epsilon \cdot \sigma_z$ (where $\epsilon \sim \mathcal{N}(0, \sigma^2)$), $\phi\text{-QFT}$ with Bell corrections has:*

$$|peak|_{\phi\text{-QFT}} \propto \exp\left(-\frac{\xi\epsilon^2}{\Delta f}\right) > 1 - \epsilon \propto |peak|_{\text{standard}} \quad (22)$$

for $\epsilon < 0.1$. The damping factor itself is $\exp(-\xi\epsilon^2/\Delta f) < 1$; the advantage exists only in comparison with the linear degradation $1 - \epsilon$.

Proof. Standard QFT under phase noise: $|peak| \rightarrow |peak| \cdot (1 - \epsilon)$ (linear degradation).

Bell-corrected $\phi\text{-QFT}$: $|peak| \rightarrow |peak| \cdot \exp(-\xi\epsilon^2/\Delta f)$ (quadratic in ϵ).

For small ϵ :

$$e^{-\xi\epsilon^2/\Delta f} \approx 1 - \frac{\xi\epsilon^2}{\Delta f} > 1 - \epsilon \quad (23)$$

since $\xi\epsilon/\Delta f \ll 1$ for realistic $\epsilon < 0.1$. $\square$

.4 Bell Test Modifications

T0 Correlation Function

Definition .4.1 (T0 Bell Correlation). For two qubits with measurement angles a and b , the T0-modified correlation is:

$$E^{T0}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j)) \quad (24)$$

where $f(n, l, j) = (n/\phi_{\text{par}})^l \cdot (1 + \xi j/\pi)$ for quantum numbers (n, l, j) . Here $f(n, l, j)$ is the quantum-number factor in the damping term, not a frequency and not the base winding number $f = 1/\xi$.

For photon-like qubits ($n = 1, l = 0, j = 1$):

$$f(1, 0, 1) = \phi_{\text{par}}^0 \cdot \left(1 + \frac{\xi}{\pi}\right) \approx 1.000042 \quad (25)$$

CHSH Inequality Modification

Proposition .4.2 (T0 CHSH Value). For n entangled qubits, the CHSH parameter is:

$$\text{CHSH}^{T0}(n) = 2\sqrt{2} \cdot \exp\left(-\frac{\xi \ln(n)}{\Delta f}\right) \quad (26)$$

Proof. Standard CHSH for singlet state:

$$\text{CHSH}^{\text{QM}} = |E(0^\circ, 45^\circ) - E(0^\circ, 135^\circ) + E(90^\circ, 45^\circ) + E(90^\circ, 135^\circ)| = 2\sqrt{2} \quad (27)$$

With $E = -\cos(a - b)$, the value $2\sqrt{2}$ belongs to the angles $a = 0^\circ, 90^\circ$, $b = 45^\circ, 135^\circ$ (as in the appendix); the angles $0^\circ/45^\circ/22.5^\circ/67.5^\circ$ apply to the polarisation correlator $\cos 2(a - b)$.

With T0 modification from Eq. (24) and n -qubit Bell damping:

$$E_i^{\text{T0}} = E_i^{\text{QM}} \cdot (1 - \xi f(n, l, j)) \cdot e^{-\xi \ln(n)/\Delta f} \quad (28)$$

$$\approx E_i^{\text{QM}} \cdot (1 - \xi f(n, l, j)) \left(1 - \frac{\xi \ln(n)}{\Delta f}\right) \quad (29)$$

The factor $(1 - \xi f)$ is not negligible: $\xi f = 1.33 \times 10^{-4}$ is larger than $\xi \ln(n)/\Delta f$ at $n = 2$ (3.1×10^{-5}) and comparable at $n = 73$ (1.9×10^{-4}). Eq. (26), the summation below and the numerical values of this document do not contain this factor; with it, $\text{CHSH}^{\text{T0}}(73) = 2.827511$ instead of 2.827888. Compared with the measured $S = 2.74$ the difference cannot be resolved.

Summing over the four CHSH terms:

$$\text{CHSH}^{\text{T0}}(n) = \text{CHSH}^{\text{QM}} \cdot \left(1 - \frac{\xi \ln(n)}{\Delta f}\right) \approx 2\sqrt{2} \cdot e^{-\xi \ln(n)/\Delta f} \quad (30)$$

□

Experimental Predictions

73-Qubit Prediction

For the 73-qubit quantum "liar detector" experiment:

$$\text{CHSH}^{\text{QM}} = 2.828427 \quad (31)$$

$$\text{CHSH}^{\text{T0}}(73) = 2.828427 \cdot e^{-1.333 \times 10^{-4} \cdot 4.290/2.9999} \quad (32)$$

$$= 2.827888 \quad (33)$$

Deviation: $\Delta = 5.39 \times 10^{-4}$ (measurable only with $\sigma = 10^{-4}$ and device-stable precision; not resolvable with present NISQ hardware, Section .7). Without the factor $(1 - \xi f)$; with it, $\text{CHSH}^{\text{T0}}(73) = 2.827511$.

Table 3: T0 CHSH predictions for multi-qubit systems (column "Testable": with error-corrected hardware; none of the values is resolvable with present NISQ hardware)

n qubits	QM CHSH	T0 CHSH	Δ (%)	Testable
2	2.828427	2.828340	0.0031	Marginal
5	2.828427	2.828225	0.0072	Marginal
10	2.828427	2.828138	0.0102	Yes
20	2.828427	2.828051	0.0133	Yes
50	2.828427	2.827935	0.0174	Yes
73	2.828427	2.827888	0.0191	Yes
100	2.828427	2.827848	0.0205	Yes

Spatial Correlation Delay

Proposition .4.3 (Ansatz: Spatial Bell Delay). *For a Bell test over distance d , the following ansatz would give a delay:*

$$\Delta t = \xi \cdot \frac{d}{c} \quad (34)$$

Proof. The correlation field propagates causally at speed c . The T0 modification introduces a phase delay proportional to ξ :

$$\phi_{T0}(d, t) = \phi_{QM}(d, t - \Delta t) \quad (35)$$

where $\Delta t = \xi d/c$ is meant to ensure causal consistency. $\square$

This ansatz does not follow from the framework of this work: a correlation propagating causally at c and delayed by $\xi d/c$ would be a local mechanism. The fractal damping, however, establishes no local bound, and the Bell violation remains (Section .4). The delay is therefore not listed as a prediction of FFGFT.

Satellite Test

For $d = 1000$ km:

$$\Delta t = 1.333 \times 10^{-4} \times \frac{1000 \text{ km}}{299792 \text{ km/s}} = 444.75 \text{ ns} \quad (36)$$

A delay of this size would be resolvable with atomic clocks (precision ~ 10 ns); since the ansatz does not follow from the framework (see above), it is not a testable FFGFT deviation.

.5 Application to Shor's Algorithm

Standard Shor Algorithm

Shor's algorithm factors N by finding the period r of the function $f(x) = a^x \bmod N$:

Algorithm 1 Standard Shor Algorithm

- 1: Choose random $a \in [2, N - 1]$ with $\gcd(a, N) = 1$
 - 2: Initialize $|\psi_0\rangle = |0\rangle^{\otimes n}$
 - 3: Apply Hadamard: $|\psi_1\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$
 - 4: Compute $f(x)$: $|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |a^x \bmod N\rangle$
 - 5: Measure second register, collapse to $|\psi_3\rangle = \frac{1}{\sqrt{2^n/r}} \sum_{k=0}^{2^n/r-1} |kr\rangle$
 - 6: Apply QFT: $|\psi_4\rangle = \text{QFT}|\psi_3\rangle$
 - 7: Measure, obtain $y \approx 2^n \cdot s/r$
 - 8: Extract r via continued fractions
 - 9: Compute factors: $\gcd(a^{r/2} \pm 1, N)$
-

T0-Shor with ϕ -QFT

Algorithm 2 T0-Shor Algorithm

- 1: Choose random a with $\gcd(a, N) = 1$
 - 2: Initialize T0 qubits with ϕ -hierarchy: $\theta_k = 2\pi/\phi_{\text{par}}^k$
 - 3: Apply Bell-damped Hadamard: $H_{\text{T0}}^{(n)}$ (Eq. 9)
 - 4: **Order finding:** Test $r \in [2, 100]$ for $a^r \equiv 1 \pmod{N}$
 - 5: **if** period found **then**
 - 6: **return** period r
 - 7: **end if**
 - 8: ϕ -hierarchy search: Test $r = \text{round}(\phi_{\text{par}}^k)$ for $k \in [0, 20]$
 - 9: **if** $a^r \equiv 1 \pmod{N}$ **then**
 - 10: **return** period r
 - 11: **end if**
 - 12: Apply ϕ -QFT with Bell corrections
 - 13: Deterministic measurement (energy field readout)
 - 14: Extract r via continued fractions
 - 15: Compute factors
-

Complexity Analysis

Proposition .5.1 (T0-Shor Complexity). *The T0-Shor algorithm with preceding order finding has complexity:*

$$\mathcal{O}(\log^3 N) \quad (37)$$

The preceding order finding costs $\mathcal{O}(\log^2 N)$ and does not change the complexity (Appendix 0.2); an additional term in ξ follows from no step.

.6 Structural Placement

[K] Independently of the hardware question there is a structural limit for resonance-based methods. By Weyl's law the eigenvalue density of any compact state space – and an n -qubit register is one – grows by a power law $N(\lambda) \sim \lambda^{d/2}$. Arithmetic counting densities, by contrast, grow as $T \ln T$ (Riemann-von Mangoldt). That is not a power; a finite state space cannot carry such a spectrum in full (Doc. 316, August 2026).

[K] Preparation (modular exponentiation) is the real problem and accounts for approximately **95 %** of the gate cost; the quantum Fourier transform for only around 5 % (Doc. 176, August 2026).

[K] The quantum Fourier transform moreover generates no new information. It makes visible periodicities already present in the prepared states. The limiting step in Shor's algorithm is state preparation – the evaluation of $a^x \bmod N$ – not the transform itself.

[K] The resonance structure of period finding rests on primes. The bicoherence test on 2469 Riemann zeros (Doc. 316) shows phase coupling exactly at prime powers and none at composite numbers. The geometry parameter $\xi = 4/30000$ has no function in $\mathbb{Z}/N\mathbb{Z}$; the period r is determined by $\text{lcm}(p-1, q-1)$.

.7 Experimental Validation with IBM Quantum Hardware

Hardware Tests on 73-Qubit and 127-Qubit Systems

Initial validation runs were performed on IBM Quantum (Brisbane, Sherbrooke; 127 physical qubits each) during 2025. An extended, statistically robust repetition was carried out on 28 May 2026 on `ibm_kingston` (Heron r2, 156 qubits) with 50 independent runs; the values in this section refer to that 50-run measurement.

Bell State Fidelity Tests

Bell State Generation Protocol

Circuit: Standard Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$

- Apply Hadamard gate to qubit 0
- Apply CNOT with control=0, target=1
- Measure both qubits
- Repeat for 2048 shots, 50 independent runs

Results from 50 independent runs on `ibm_kingston` (28 May 2026):

Table 4: Bell state fidelity: Experimental results (50 runs)

Quantity	$P(00\rangle)$	$P(11\rangle)$	Leakage	Fidelity
Mean	0.5091	0.4785	0.0124	0.9876
Std dev	0.0112	0.0109	—	0.0028

Statistical Analysis:

$$\text{Mean fidelity} = 0.9876 \pm 0.0028 \quad (38)$$

$$\text{Variance } P(|00\rangle) = 1.249 \times 10^{-4} \quad (39)$$

$$\text{Shot-noise expectation} = 0.25/2048 = 1.221 \times 10^{-4} \quad (40)$$

Comparison with standard QM expectation:

- Ratio of observed variance to shot-noise: 1.02
- The repeatability variance is **statistically consistent** with ordinary quantum shot-noise.

Important

Finding: The 50-run measurement shows a variance indistinguishable from shot-noise (ratio 1.02); a deterministic variance reduction relative to QM does not exist. The high Bell fidelity is confirmed.

Chi-square test for repeatability

Null hypothesis: variance of $P(|00\rangle)$ equals the shot-noise variance.

$$\chi^2 = (n-1)s^2/\sigma_0^2 = 50.14, \quad \text{df} = 49, \quad p = 0.856 \quad (41)$$

Conclusion: $p \gg 0.05 \Rightarrow$ the observed variance is fully consistent with pure shot-noise. No evidence of sub-shot-noise determinism.

CHSH Parameter Measurements

On 28 May 2026 a real CHSH-parameter measurement was carried out on hardware (15 measurements on `ibm_kingston`, 10 on `ibm_marrakesh`, four angle settings each, 2048 shots). It is the first hardware-measured CHSH determination of this corpus, fully documented with job IDs.

Measurement setup

Alice angles $a = 0^\circ$, $a' = 45^\circ$; Bob $b = 22.5^\circ$, $b' = -22.5^\circ$. Correlator $E = [N_{00} - N_{01} - N_{10} + N_{11}]/N$. CHSH combination $S = E(a, b) + E(a, b') + E(a', b) - E(a', b')$.

Table 5: CHSH measurement: Hardware results (28 May 2026)

Device	Measurements	Mean S	SE	% Tsirelson	Bell violation
<code>ibm_kingston</code>	15	2.7396	0.0071	96.9%	105σ
<code>ibm_marrakesh</code>	10	2.7016	0.0138	95.5%	51σ
pooled	25	2.7244	0.0078	96.3%	—

Reference values: Tsirelson bound $2\sqrt{2} = 2.8284$; classical bound 2.0.

Bell violation confirmed

The measured CHSH value $S = 2.74$ lies 105σ (Kingston) above the classical bound. Local realism is excluded; the entanglement generated is genuine. At 96.9% of the Tsirelson bound this is an excellent result for NISQ hardware.

Deviation from the Tsirelson bound

The measured value lies 3.14% below $2\sqrt{2}$. The mean correlator magnitude is 0.685 instead of the ideal 0.707 — corresponding to $\sim 1.5\%$ gate and readout error (NISQ characteristic). This deviation is **decoherence, not the ξ -effect**: the predicted ξ -effect is $\sim 10^{-5}$ (order 0.001% of S), about $3000\times$ smaller than the measured gap.

Important

Data basis: The only substantiated CHSH determination is the direct measurement of 28 May 2026 ($S = 2.74$). The 2025 hardware validation run concerned only Bell-state generation, not the CHSH parameter.

Cross-platform finding on the measurability of ξ

Two identical-model Heron r2 chips differ in their measured CHSH value by 0.038 ($\approx 1.4\%$ of S ; Welch t -test $t = 2.46$, $p = 0.028$), purely through calibration and noise differences. Since the sought ξ -effect is $\sim 1400\times$ smaller than this device-to-device variation, the ξ -prediction is **in principle not resolvable** with present NISQ hardware — a directly measured, not merely estimated, finding. The ξ -test requires error-corrected qubits with sub-ppm, device-stable precision.

73-Qubit Bell Test

Apparatus: error-corrected quantum processor; with NISQ processors such as IBM Eagle or Heron and Google Sycamore the deviation is not resolvable (Section .7)

Protocol:

1. Prepare 73-qubit GHZ state: $|\text{GHZ}_{73}\rangle = (|0\rangle^{\otimes 73} + |1\rangle^{\otimes 73})/\sqrt{2}$
2. Apply measurement angles: $a \in \{0^\circ, 90^\circ\}$, $b \in \{45^\circ, 135^\circ\}$ (for $E = -\cos(a - b)$; the angles $0^\circ/22.5^\circ/45^\circ/67.5^\circ$ apply only to the polarisation correlator $\cos 2(a - b)$)
3. Compute pairwise correlations $E(a_i, b_j)$ for all pairs
4. Compute $\text{CHSH} = \sum_i E(a_i, b_i) - E(a_i, b_{i+1})$
5. Repeat 10^6 times, compute mean and standard error
6. Compare with predictions (Table 3)

Expected result:

$$\text{CHSH}_{\text{measured}} = 2.8279 \pm 0.0001 \quad (42)$$

Falsification criteria:

- If $\text{CHSH}_{\text{measured}} = 2.8284 \pm 0.0001$: T0 falsified
- If $\text{CHSH}_{\text{measured}} = 2.8279 \pm 0.0001$: T0 confirmed (5σ)

Satellite Bell Test

Apparatus: Micius satellite or future ESA quantum link

Protocol:

1. Generate entangled photon pairs at satellite
2. Send to ground stations A and B ($d = 1000$ km apart)
3. Synchronize via atomic clocks (GPS, precision ~ 10 ns)
4. Measure correlation arrival times with femtosecond lasers
5. Compare timestamps: $\Delta t_{AB} = t_B - t_A - d/c$

Classification: The protocol tests the ansatz of the spatial correlation delay $\Delta t = \xi d/c \approx 445$ ns (Section .4), which does not follow from the framework of this work. It is therefore not a falsification test of FFGFT.

.8 Implementation and Results

Python Implementation

We provide two implementations:

1. Complete theoretical implementation (630 lines):

- Complete T0 qubit class with energy field dynamics
- ϕ -QFT with Bell corrections
- Bell-corrected entanglement damping
- Deterministic measurement via field readout

2. Production hybrid implementation (400 lines):

- Order finding (classical test of r for $a^r \equiv 1 \pmod{N}$)
- ϕ -hierarchy search
- Classical fallback for robustness

- Complete benchmark suite

Benchmark Results

Table 6: T0-Shor performance on benchmark suite

N	Factors	Period r	Method	Time (s)	Success
15	3×5	4	Order finding	0.033	✓
21	3×7	2	Order finding	0.0003	✓
33	3×11	10	Order finding	0.0003	✓
35	5×7	12	Order finding	0.0002	✓
77	7×11	30	Order finding	0.0003	✓
143	11×13	60	Order finding	0.0003	✓
Success rate: 6/6 (100 percent)					

Code Snippet: Order Finding

```
def find_period_xi_resonance(self, a: int) -> Optional[int]:
    '''Uses T0 energy field resonances'''
    best_r = None
    max_resonance = 0

    for r in range(2, min(self.N, 100)):
        # Energy signature
        power = pow(a, r, self.N)

        # T0 fractal damping
        xi_modulation = np.exp(-XI * r * r / DF)

        # Resonance at  $a^r \equiv 1 \pmod{N}$ 
        resonance_strength = xi_modulation / (abs(power - 1) + 1)

        if abs(power - 1) < 0.01:
            return r # Strong resonance

    return best_r
```

The factor xi_modulation is strictly decreasing in r and does not shift the result; the period is determined solely by the condition $a^r \equiv 1 \pmod{N}$. The procedure is a classical order finding.

.9 Discussion

Theoretical Implications

1. **Determinism restored:** Energy field qubits provide deterministic framework compatible with quantum interference
2. **No local bound:** The fractal damping changes CHSH only far below the measurement noise; it establishes no local bound, and the Bell violation remains

3. **Measurement problem solved:** Measurement is field readout, not probabilistic collapse
4. **Improved stability:** ξ -damping provides natural decoherence suppression

Experimental Testability

The CHSH predictions (73-qubit test, CHSH scaling) are not resolvable with present NISQ hardware (Section .7); they require error-corrected qubits. In the QM sector there is currently no FFGFT deviation testable with present hardware.

Limitations and Open Questions

1. **Scalability:** Tested up to $N = 143$; RSA-2048 requires further analysis
2. **Hardware implementation:** Requires specialized qubit frequencies (ϕ -hierarchy)
3. **Quantum error correction:** Integration with surface codes remains open
4. **Many-body systems:** Extension to > 100 qubits needs refinement

.10 Conclusion

We have presented a theoretical framework and a hardware test of quantum computing within the T0 time-mass duality framework. The main contributions are:

Theoretical Achievements

1. Mathematical framework

- Proof attempt for the ϕ -QFT equivalence to standard QFT for period finding (Theorem .3.4); not proven in its present form
- Bell-corrected entanglement framework with predictions that are testable only with error-corrected hardware
- Demonstration of improved stability through fractal damping (Corollary .3.5)

2. New physical insights

- Energy field qubits provide deterministic alternative to probabilistic collapse
- Correlation fields describe the Bell correlations; the fractal damping establishes no local bound, and the Bell violation remains
- ξ -damping acts as natural decoherence suppression

Experimental Validation

3. IBM Quantum hardware tests (2025 + May 2026)

- **Bell fidelity:** 98.8 percent (50 runs, ibm_kingston, May 2026); variance consistent with shot-noise (ratio 1.02), no variance reduction relative to QM
- **CHSH (measured):** $S = 2.74$ (96.9 percent of the Tsirelson bound), clear Bell violation (105σ over classical)
- **ξ measurability:** device-to-device variation (≈ 1.4 percent) exceeds the ξ -effect by $\sim 1400\times$; not resolvable with NISQ hardware
- **Runnability:** T0 circuits execute cleanly on production hardware

4. Monte Carlo validation

- 10,000-run simulations are compatible with the IBM measured values ($p = 0.204$); since the ξ -effect lies far below the hardware noise, this is no confirmation of the T0 prediction
- The implementation accurately reproduces T0 predictions
- Bootstrap analysis provides rigorous uncertainty quantification

Key Results

Central Result

The T0 framework successfully reproduces quantum computing phenomena while offering:

1. **Deterministic foundation:** Quantum behavior emerges from energy field dynamics
2. **Bell violation remains:** The fractal damping establishes no local bound; local realism is excluded by the measurement
3. **Improved stability:** ξ -damping suppresses decoherence quadratically
4. **Experimental validation:** IBM tests confirm the Bell violation ($S = 2.74$, 3.1 percent below $2\sqrt{2}$); a confirmation of the ξ prediction to 0.02 percent does not exist
5. **Falsifiability:** Clear experimental criteria for verification/falsification

No Hardware-Fitted ξ

A difference between $\xi_{\text{base}} = 1.33 \times 10^{-4}$ and a ξ fitted to hardware values does not exist: the only substantiated CHSH measurement ($S = 2.74$, Section .7) does not resolve the ξ -effect. A hardware-fitted ξ_{fit} and a decomposition $\xi_{\text{eff}}(N) = \xi_{\text{base}} + \xi_{\text{hardware}}(N)$ with N -scaling are therefore not justified.

Implications for Quantum Computing

Practical applications:

- **Error correction:** T0-aware protocols could leverage ξ -damping for natural error suppression
- **Hardware design:** Optimize qubit frequencies to ϕ -harmonic resonances (with $E_0 = 7.398$ MeV: 5.50 GHz for $n = 9$, 2.10 GHz for $n = 10$)
- **Algorithm development:** T0-native algorithms leverage deterministic evolution
- **Benchmarking:** ξ parameter as quality metric for quantum processors

Fundamental physics:

- Resolution of measurement problem via energy field readout
- Connection to broader frameworks (Causal Fermion Systems, deterministic QFT)
- Testable predictions for Planck-scale physics

Falsification Criteria

The T0 framework makes precise, falsifiable predictions:

Critical tests for T0 theory

Test 1: CHSH scaling

- Measure CHSH for $N = 10, 20, 50, 100, 200$ qubits
- T0 predicts: $S(N) = 2\sqrt{2} \cdot \exp(-\xi \ln(N)/D_f)$
- **Falsified if:** Systematic deviation $> 3\sigma$ from scaling law

Test 2: Spatial correlation delay

- Satellite Bell test over $d = 1000$ km
- Not an FFGFT criterion: the delay $\Delta t = \xi d/c \approx 445$ ns presupposes a local propagation of the correlations and does not follow from the framework (Section .4)

Test 3: Bell fidelity variance

- Measure variance over 100+ runs on same system
- T0 predicts: $\sigma^2 < 0.001$; with 2048 shots the QM shot-noise variance is at most $0.25/2048 = 1.2 \times 10^{-4}$ and thus also below this value, so the criterion does not distinguish T0 from QM
- **Falsified if:** $\sigma^2 > 0.005$ (the QM expectation is $\leq 1.2 \times 10^{-4}$); the measured variance is consistent with shot-noise (ratio 1.02, Section .7)

Test 4: ϕ -harmonic resonances

- Test qubit performance at frequencies $f_n = (E_0/h)\xi^2\phi^{-2n}$ with $E_0 = 7.398$ MeV
- T0 predicts: Reduced phase noise at 5.50 GHz ($n = 9$), 2.10 GHz ($n = 10$)
- **Falsified if:** No measurable improvement at predicted frequencies

Concluding Remarks

The T0 time-mass duality framework offers a new view of quantum computing. By replacing probabilistic amplitudes with deterministic energy fields, we achieve:

- **Conceptual clarity:** No measurement paradox, no wavefunction collapse
- **Mathematical framework:** Equivalence with standard quantum algorithms formulated as a proof attempt, not proven in its present form
- **Experimental support:** IBM tests confirm the Bell violation ($S = 2.74$); a validation of the ξ prediction to 0.02 percent does not exist
- **Practical utility:** Natural error suppression and hardware optimization strategies
- **Falsifiability:** Clear criteria for experimental verification/falsification

While extraordinary claims require extraordinary evidence, the theoretical framework, the proof attempt and the hardware measurement of the Bell violation presented here provide a reason for serious consideration of the T0 framework; the characteristic ξ prediction is not testable with present hardware.

We invite the quantum computing community to:

1. **Replicate** our experimental protocols on independent hardware
2. **Scrutinize** our theoretical derivations and identify potential errors
3. **Extend** the framework to new domains (quantum chemistry, many-body physics)
4. **Test** the falsification criteria with high-precision experiments
5. **Collaborate** on developing T0-optimized quantum technologies

The decisive test of any physical theory is its ability to predict, explain, and unify phenomena. The T0 framework, as demonstrated in this work, shows promise on all three

fronts. Whether it represents a fundamental truth about nature or remains a useful effective description must be determined through rigorous experimental testing and theoretical development.

The test of all knowledge is experiment. Experiment is the sole judge of scientific 'truth'.
— Richard Feynman

We look forward to the experimental tests that will ultimately validate or falsify this framework, and commit to transparently updating our conclusions based on empirical evidence.

.11 Cosmic Corrections for Quantum Computing

The Wrinkled Torus Universe and Qubits

Based on the insight that the universe is a wrinkled fractal torus ($\Delta f = 3 - \xi$), it follows that qubits are local manifestations of this universal geometry. This has concrete implications for quantum computing.

Implementation of Cosmic Corrections

For practical implementation, we have developed three Python scripts:

1. **T0 Cosmic Qubit Simulator:**
2/python/t0_cosmic_qubit_simulator.py
Simulates qubits in the wrinkled universe with day, moon, and annual cycle corrections.
2. **T0 Cosmic Error Correction:**
2/python/t0_cosmic_error_correction.py
Implements cosmically synchronous error correction for quantum gates and optimal start times.
3. **T0 Cosmic Data Analyzer:**
2/python/t0_cosmic_data_analyzer.py
Analyzes IBM Quantum data for periodic cosmic signatures and positional correlations.

Experimental Tests

The scripts enable the following experimental tests with IBM Quantum hardware:

- **Position dependence:** Qubits on gyri vs. sulci of the chip geometry
- **Time dependence:** 24-hour and 12-hour periods in qubit performance
- **Fractal arrangement:** Optimized qubit placement based on Δf

Expected Improvements

By applying the cosmic corrections, we expect:

- 10-20 percent longer coherence times T_2
- 5-15 percent reduced gate errors
- Optimal algorithm performance at specific cosmic times

All scripts are available in the GitHub repository:

<https://github.com/jpascher/T0-Time-Mass-Duality/tree/main/2/python/>

Data Availability

Complete Python implementation available at:

<https://github.com/jpascher/T0-Time-Mass-Duality>

All experimental protocols and benchmark data are provided in the supplementary materials.

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 Implements cosmically synchronous error correction for quantum gates and optimal start times.
3. **T0 Cosmic Data Analyzer:**
`2/python/t0_cosmic_data_analyzer.py`

External dependencies: The three scripts require numpy, scipy, matplotlib, pandas and ephemeris. A `requirements_147.txt` is located at `2/python/requirements_147.txt`; installation: `pip install -r requirements_147.txt`. matplotlib is optional in the first two scripts (for graphics only).

Analyzes IBM Quantum data for periodic cosmic signatures and positional correlations.

Experimental Tests

The scripts enable the following experimental tests with IBM Quantum hardware:

- **Position Dependence:** Qubits on gyri vs. sulci of the chip geometry
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0.1 Detailed Proofs

Proof of Lemma .3.2

Proof attempt for Lemma .3.2: For every period $r \in [2, N]$ with $N < 2^{20}$, there exists $k \in \mathbb{Z}$ and rational number c with small denominator such that $|r - \phi_{\text{par}}^k \cdot c| < 1/(2r^2)$.

Step 1: Irrational distribution of ϕ_{par} powers. The golden ratio $\phi_{\text{par}} = (1 + \sqrt{5})/2$ is a Pisot number with minimal polynomial $x^2 - x - 1 = 0$. By the three-dimensional Weyl equidistribution theorem, the triples

$$\left(\left\{ \frac{\phi_{\text{par}}^k}{r} \right\}, \left\{ \frac{\phi_{\text{par}}^{k+1}}{r} \right\}, \left\{ \frac{\phi_{\text{par}}^{k+2}}{r} \right\} \right)$$

for $k = 0, 1, \dots, K$ are uniformly distributed in the unit cube $[0, 1)^3$, since ϕ_{par} , ϕ_{par}^2 , and ϕ_{par}^3 are linearly independent over $\mathbb{Q}$.

Step 2: Diophantine approximation. For each $r \in [2, N]$, consider the sequence $\{\phi_{\text{par}}^k \bmod r\}$ for $k = 0, \dots, \lceil \log_{\phi_{\text{par}}}(2r^2) \rceil$. Since the sequence is uniformly distributed, by the Dirichlet pigeonhole principle there exist $k_1 < k_2$ such that:

$$|\phi_{\text{par}}^{k_1} - \phi_{\text{par}}^{k_2}| \bmod r < \frac{r}{M}$$

where $M = \lceil \log_{\phi_{\text{par}}}(2r^2) \rceil + 1$.

Step 3: Construction of approximation. Let $d = k_2 - k_1$. Then:

$$\phi_{\text{par}}^{k_1} \cdot (\phi_{\text{par}}^d - 1) = m \cdot r + \epsilon$$

with $|\epsilon| < r/M$, where $m \in \mathbb{Z}$. Rearranging gives:

$$r = \frac{\phi_{\text{par}}^{k_1}}{m} \cdot (\phi_{\text{par}}^d - 1) - \frac{\epsilon}{m}$$

Set $c = (\phi_{\text{par}}^d - 1)/m$. Since ϕ_{par}^d is integral up to a Fibonacci recurrence, m is small. In particular, for $d = 1, 2, 3, 4$:

$$\begin{aligned} \phi_{\text{par}}^1 - 1 &= 0.618 \approx \frac{5}{8} \\ \phi_{\text{par}}^2 - 1 &= 1.618 \approx \frac{13}{8} \\ \phi_{\text{par}}^3 - 1 &= 3.236 \approx \frac{26}{8} \\ \phi_{\text{par}}^4 - 1 &= 5.854 \approx \frac{47}{8} \end{aligned}$$

Step 4: Error estimate. With $M > 2r^2$ and $m \leq r$ (since $\phi_{\text{par}}^{k_1} < r^2$), we obtain:

$$|r - \phi_{\text{par}}^{k_1} \cdot c| = \left| \frac{\epsilon}{m} \right| < \frac{r/M}{1} < \frac{1}{2r^2}$$

Step 5: Bound for $N < 2^{20}$. For $N < 2^{20}$, $\log_{\phi_{\text{par}}}(N) < \frac{20}{\log_2(\phi_{\text{par}})} \approx 28.8$. Thus k values up to 29 suffice. The computed approximations:

$$\begin{aligned} r = 2 : \quad \phi_{\text{par}}^1 &= 1.618, \quad c = 1.236, \quad \text{error} = 0.0001 \\ r = 3 : \quad \phi_{\text{par}}^2 &= 2.618, \quad c = 1, \quad \text{error} = 0.382 \\ r = 4 : \quad \phi_{\text{par}}^3 &= 4.236, \quad c = 1, \quad \text{error} = 0.236 \\ r = 5 : \quad \phi_{\text{par}}^4 &= 6.854, \quad c = 0.729, \quad \text{error} = 0.005 \end{aligned}$$

Since $1/(2r^2) \leq 1/8 = 0.125$ for $r \geq 2$, the errors 0.382 ($r = 3$) and 0.236 ($r = 4$) do not satisfy the condition $< 1/(2r^2)$. The appendix proof of the lemma therefore does not hold. At $r = 2$, $\phi_{\text{par}} \cdot 1.236 = 1.9999$, so the error is ≈ 0.0001 .

Proof of Theorem .3.4

Proof attempt:

Part A: Signal analysis Let $f(x) = a^x \bmod N$ with period r . After measuring the function register in the standard Shor algorithm, we obtain:

$$|\psi_3\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |jr + \ell\rangle$$

where $M = \lfloor Q/r \rfloor$ and $\ell \in [0, r-1]$ is random.

The QFT yields:

$$|\psi_4\rangle = \frac{1}{\sqrt{QM}} \sum_{y=0}^{Q-1} \sum_{j=0}^{M-1} e^{2\pi i(jr+\ell)y/Q} |y\rangle$$

The amplitude at y is:

$$\alpha(y) = \frac{1}{\sqrt{QM}} e^{2\pi i \ell y/Q} \sum_{j=0}^{M-1} e^{2\pi i j r y/Q}$$

Part B: ϕ -QFT modification For ϕ -QFT, we replace $Q = 2^n$ with $Q_{\phi_{\text{par}}} = \phi_{\text{par}}^n$ and obtain:

$$\alpha_{\phi}(y) = \frac{1}{\sqrt{Q_{\phi_{\text{par}}} M_{\phi}}} e^{2\pi i \ell y/Q_{\phi_{\text{par}}}} \sum_{j=0}^{M_{\phi}-1} e^{2\pi i j r y/Q_{\phi_{\text{par}}}}$$

with $M_{\phi} = \lfloor Q_{\phi_{\text{par}}}/r \rfloor$.

The phase $\theta = 2\pi j r y/Q_{\phi_{\text{par}}}$ is modified by Bell damping:

$$\tilde{\alpha}_{\phi}(y) = \alpha_{\phi}(y) \cdot \exp\left(-\xi \frac{\theta^2}{\pi^2 \Delta f}\right)$$

Part C: Peak positions The main peaks occur when $r y/Q_{\phi_{\text{par}}}$ is close to an integer s :

$$y_{\text{peak}} \approx \frac{s \cdot Q_{\phi_{\text{par}}}}{r}$$

For standard QFT: $y_{\text{peak}} \approx s \cdot 2^n/r$ For ϕ -QFT: $y_{\text{peak}} \approx s \cdot \phi_{\text{par}}^n/r$

Part D: Error analysis The maximum phase error at a peak is:

$$\Delta\phi = 2\pi \left(\frac{r y}{Q_{\phi_{\text{par}}}} - s \right)$$

By Lemma .3.2, there exists k such that:

$$\left| \frac{Q_{\phi_{\text{par}}}}{r} - \frac{2^n}{r} \cdot \frac{\phi_{\text{par}}^k}{2^k} \right| < \frac{0.2 \cdot 2^n}{r}$$

Therefore:

$$\left| y_{\phi} - y_{\text{std}} \cdot \frac{\phi_{\text{par}}^k}{2^k} \right| < 0.2 y_{\text{std}}$$

Part E: Continued fraction stability The continued fraction expansion extracts s/r from y/Q if:

$$\left| \frac{y}{Q} - \frac{s}{r} \right| < \frac{1}{2r^2}$$

Our error is:

$$\left| \frac{y_{\phi}}{Q_{\phi_{\text{par}}}} - \frac{y_{\text{std}}}{2^n} \cdot \frac{\phi_{\text{par}}^k}{2^k} \right| < \frac{0.2}{r}$$

With $\frac{\phi_{\text{par}}^k}{2^k} \approx 1$ for optimal k , $0.2/r < 1/(2r^2)$ would be required; this holds only for $r < 2.5$, i.e. only for $r = 2$. For general r the continued fraction condition is therefore not established. Example $r = 3$: $0.067 > 0.056$.

Part F: Success probability The success probability for standard Shor is:

$$P_{\text{std}} = \frac{4}{\pi^2} - \frac{1}{3r} + O(r^{-2})$$

For ϕ -QFT with Bell damping:

$$P_{\phi} = P_{\text{std}} \cdot \left(1 - \frac{\xi \ln(r)}{\Delta f} \right) + \Delta P$$

$$\Delta P = \frac{\xi}{\pi^2} \cdot \frac{\sin^2(\pi r/2)}{r^2}$$

Since $\xi \ln(r)/\Delta f \sim 10^{-4}$ and $\Delta P \sim \xi/r^2$, we have:

$$P_{\text{std}} \leq P_{\phi} \leq P_{\text{std}} + \xi$$

□

0.2 Implementation Details

Monte Carlo Simulation for Bell Tests

The complete algorithm for the Monte Carlo simulation of 73-qubit Bell tests:

Algorithm 3 Monte Carlo Bell Test Simulation**Require:** ξ : T0 coupling parameter, n : number of qubits, N_{runs} : simulations**Ensure:** CHSH mean, standard error, distribution

```

1: Initialize  $\Delta f = 3 - \xi$ 
2: Define measurement angles:  $\theta = [(0, \pi/4), (0, 3\pi/4), (\pi/2, \pi/4), (\pi/2, 3\pi/4)]$ 
3: Initialize chsh_values = []
4: for  $i = 1$  to  $N_{\text{runs}}$  do
5:   correlations = []
6:   for  $(a, b)$  in  $\theta$  do
7:      $\Delta\theta = a - b$ 
8:     damping =  $\exp(-\xi \cdot \ln(n)/\Delta f)$ 
9:      $E = -\cos(\Delta\theta) \cdot \text{damping}$  {Singlet: negative sign}
10:    correlations.append( $E$ )
11:   end for
12:   chsh = |correlations[0] - correlations[1] + correlations[2] + correlations[3]|
13:   Add shot noise:  $\text{chsh} \leftarrow \text{chsh} + \mathcal{N}(0, 1/\sqrt{\text{shots}})$ 
14:   Add field fluctuations:  $\text{chsh} \leftarrow \text{chsh} + \mathcal{N}(0, \xi^2 \cdot 0.1)$ 
15:   chsh_values.append(chsh)
16: end for
17: Compute mean  $\mu = \text{mean}(\text{chsh\_values})$ 
18: Compute standard deviation  $\sigma = \text{std}(\text{chsh\_values})$ 
19: Compute standard error SEM =  $\sigma/\sqrt{N_{\text{runs}}}$ 
20: return  $\{\mu, \sigma, \text{SEM}, \text{chsh\_values}\}$ 

```

Complexity Analysis of T0-Shor**Theorem:** The T0-Shor algorithm has time complexity $\mathcal{O}(\log^3 N)$.**Proof:****Step 1: Standard Shor complexity**

- Modular exponentiation: $\mathcal{O}(\log^3 N)$ via repeated squaring
- QFT: $\mathcal{O}(\log^2 N)$
- Total: $\mathcal{O}(\log^3 N)$

Step 2: T0 extensions

- Order finding: Test $r \in [2, R]$ with $R = \min(100, \sqrt{N})$
- Each test: $a^r \bmod N$ via fast exponentiation: $\mathcal{O}(\log r \cdot \log^2 N)$
- Total for scan: $\mathcal{O}(R \cdot \log R \cdot \log^2 N) = \mathcal{O}(\log^2 N)$ for constant R
- ϕ -hierarchy search: Test $k \in [0, \lceil \log_{\phi_{\text{par}}}(N) \rceil]$
- Each test: $\mathcal{O}(\log^2 N)$
- Total: $\mathcal{O}(\log N \cdot \log^2 N) = \mathcal{O}(\log^3 N)$

Step 3: Bell damping calculation For each qubit gate: multiplication by $\exp(-\xi \ln(n)/\Delta f)$

- Cost: $\mathcal{O}(1)$ per gate
- With n qubits and $\mathcal{O}(n^2)$ gates: $\mathcal{O}(n^2)$
- Since $n = \mathcal{O}(\log N)$: $\mathcal{O}(\log^2 N)$

Step 4: Total complexity

$$T_{\text{T0-Shor}}(N) = \underbrace{\mathcal{O}(\log^3 N)}_{\text{Standard Shor}} + \underbrace{\mathcal{O}(\log^2 N)}_{\text{order finding}} + \underbrace{\mathcal{O}(\log^3 N)}_{\phi\text{-search}} + \underbrace{\mathcal{O}(\log^2 N)}_{\text{Bell damping}}$$

$$= \mathcal{O}(\log^3 N)$$

A term $\mathcal{O}(\xi \log N)$ follows from no step; a constant factor ξ has no meaning inside $\mathcal{O}(\cdot)$.

Python Code Snippets

Implementation of order finding:

Listing 1: Order finding

```
def find_period_xi_resonance(a: int, N: int, max_r: int = 100) ->
    Optional[int]:
    """
    Finds period r using T0 energy field resonances.

    Args:
    a: base for modular exponentiation
    N: number to factor
    max_r: maximum period to test

    Returns:
    Period r or None if not found
    """
    XI = 4/30000 # T0 coupling constant
    D_F = 3 - XI # Fractal dimension

    best_r = None
    best_resonance = -np.inf

    for r in range(2, min(N, max_r) + 1):
        # Compute a^r mod N
        power = pow(a, r, N)

        # T0 fractal damping
        xi_modulation = np.exp(-XI * r * r / D_F)

        # Resonance strength: maximum energy at a^r ≡ 1 (mod N)
        resonance = xi_modulation / (abs(power - 1) + 1)

        # Strong resonance detected
        if abs(power - 1) < 1e-10: # Exact match
            return r

        if resonance > best_resonance:
            best_resonance = resonance
            best_r = r

        # If strong resonance (1 percent tolerance)
        if best_resonance > 100: # Strong peak
            return best_r

    return None
```

Bell damping implementation for multi-qubit systems:

Listing 2: Bell damping correction

```
class T0Qubit:
    """T0 qubit with energy field representation"""
```

```

def __init__(self, z: float, r: float, theta: float):
    """
    Args:
    z: projection onto computational basis [-1, 1]
    r: superposition amplitude [0, 1]
    theta: phase [0,  $\pi$ 2)
    """
    assert -1 ≤ z ≤ 1, f'z={z} outside [-1, 1]'
    assert 0 ≤ r ≤ 1, f'r={r} outside [0, 1]'
    assert abs(z**2 + r**2 - 1) < 1e-10, f'Norm violation:
    ↪ z2+r2={{z**2+r**2}}'

    self.z = z
    self.r = r
    self.theta = theta % (2*np.pi)
    self.XI = 4/30000
    self.D_F = 3 - self.XI

def apply_bell_damping(self, n_qubits: int):
    """
    Applies Bell damping for n-qubit system.

    Damping follows:  $\exp(-\ln(n)/D\_F)$ 
    """
    damping = np.exp(-self.XI * np.log(n_qubits) / self.D_F)
    self.z *= damping
    self.r *= damping
    # Renormalize
    norm = np.sqrt(self.z**2 + self.r**2)
    self.z /= norm
    self.r /= norm

def apply_hadamard_t0(self, n_qubits: int):
    """
    T0 Hadamard gate with Bell damping.

    Transformation: (z, r,  $\theta$ ) → (r, z,  $\theta + \pi/2$ )
    """
    # Basis change
    new_z = self.r
    new_r = self.z

    # Apply Bell damping
    self.z = new_z
    self.r = new_r
    self.apply_bell_damping(n_qubits)

    # Phase shift
    self.theta = (self.theta + np.pi/2) % (2*np.pi)

    return self

def measure_deterministic(self) -> int:
    """
    Deterministic measurement via energy field readout.

    Returns: 0 if z > 0, else 1
    """

```

```

# Energy field strength
energy_field = self.z**2 - self.r**2

if energy_field > 0:
    return 0 # >|0 state dominates
else:
    return 1 # >|1 state dominates

```

Error Analysis and Robustness

Theorem (Robustness of ϕ -QFT): Under phase noise with variance σ^2 , ϕ -QFT with Bell corrections has error rate $\mathcal{O}(\xi\sigma^2)$ compared to $\mathcal{O}(\sigma)$ for standard QFT.

Proof: Let $\epsilon \sim \mathcal{N}(0, \sigma^2)$ be phase noise. For standard QFT:

$$|\alpha_{\text{std}}(y)| \rightarrow |\alpha_{\text{std}}(y)| \cdot (1 - |\epsilon|) + \mathcal{O}(\epsilon^2)$$

For ϕ -QFT with Bell damping $\mathcal{D}(\theta) = \exp(-\xi\theta^2/(\pi^2\Delta f))$:

$$\begin{aligned}
 |\alpha_\phi(y)| &\rightarrow |\alpha_\phi(y)| \cdot \mathcal{D}(2\pi k r y / Q_{\phi_{\text{par}}} + \epsilon) \\
 &= |\alpha_\phi(y)| \cdot \exp\left(-\xi \frac{(2\pi k r y / Q_{\phi_{\text{par}}} + \epsilon)^2}{\pi^2 \Delta f}\right) \\
 &= |\alpha_\phi(y)| \cdot \left(1 - \frac{\xi \epsilon^2}{\Delta f} + \mathcal{O}(\epsilon^4)\right)
 \end{aligned}$$

Since $\xi \approx 1.333 \times 10^{-4}$, the leading error term is quadratic in ϵ , while for standard QFT it is linear.

Corollary: For $\sigma = 0.1$:

$$\begin{aligned}
 \text{Error}_{\text{std}} &\approx 10\% \\
 \text{Error}_{\phi\text{-QFT}} &\approx \frac{\xi}{\Delta f} \cdot 0.01 \approx 4.44 \times 10^{-7}
 \end{aligned}$$

In the IBM tests the measured variance is consistent with shot-noise (ratio 1.02, Section .7); a variance reduction relative to QM was not observed.

Numerical Stability and Accuracy

The implementation uses the following techniques for numerical stability:

1. **Logarithmic computation:** Instead of directly computing $\exp(-\xi \ln(n)/D_F)$, we use:

$$\text{damping} = \exp\left(-\frac{\xi}{D_F} \cdot \ln(n)\right)$$

with double precision (64-bit floats).

2. **Energy field normalization:** After each operation:

$$(z, r) \leftarrow \frac{(z, r)}{\sqrt{z^2 + r^2}}$$

3. **Phase wrapping:** Angles are always kept modulo 2π :

$$\theta \leftarrow \theta \bmod 2\pi$$

4. **Resonance detection:** Instead of exact equality $a^r \equiv 1 \pmod{N}$:

$$\text{resonance_threshold} = \max(1e-10, 1/\sqrt{N})$$

This ensures robustness even with numerical inaccuracies.